

PERSONAL INFORMATION

Christian Tamantini



📍 via dei Glicini 8, 01100, Viterbo, VT, Italy

☎ +39 392 1424159

✉ tamantinichristian@gmail.com christian.tamantini@cnr.it

🔗 <https://chritama.github.io/ChristianTamantini/index.html>

🌐 [linkedin.com/in/christian-tamantini-393425a0](https://www.linkedin.com/in/christian-tamantini-393425a0)

🆔 ORCID [0000-0001-6238-2241](https://orcid.org/0000-0001-6238-2241)

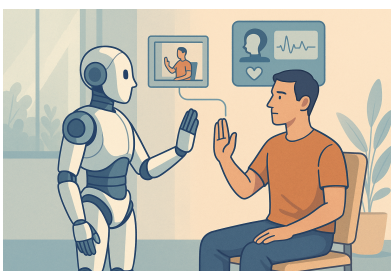
Gender Male | Date of birth 25 December 1993 | Nationality Italian

RESEARCH SUMMARY AND
CORE COMPETENCES

Researcher in robotics and cognitive human–robot interaction, with over six years of experience in the design of adaptive and user-tailored technological systems. My research advances the development of theoretical, experimental, and computational models to understand and replicate human cognitive, motor, perceptual, and emotional processes in human–machine interaction.

I have contributed to advancements in the field through the design and validation of multimodal interaction frameworks that integrate user monitoring, user-state estimation methodologies, and the generation of adaptive behavior modules. These systems combine data from visual, physiological, and kinematic signals to infer user states, such as engagement, workload, or affect, and dynamically adjust robot or system behavior to optimize interaction effectiveness and personalization. This interdisciplinary research bridges cognitive science, bioengineering, artificial intelligence, and robotics, contributing to the understanding of adaptive processes in human and artificial agents and to the development of intelligent, human-centered technologies for motor and cognitive rehabilitation.

Scientific impact is evidenced by a strong publication record (45+ Scopus-indexed papers, 367 citations, h-index 12), roles as Guest Editor and Program Chair in international conferences, and two Best Paper Awards. Overall, my work contributes to the interdisciplinary advancement of computational and experimental models of cognition and behavior, supporting the design of personalized assistive and rehabilitative technologies inspired by human psychophysiological mechanisms.



EMPLOYMENT

May 2024 – Present

Fixed-Term Researcher, Planning and Scheduling Technology Lab

Institute of Cognitive Science and Technology – ISTC-CNR, Rome, Italy

- Design and implementation of cognitive architectures for personalized, plan-based rehabilitation with real-time multimodal feedback.
- Development of user-state estimation algorithms (facial emotion recognition, workload, kinematics) for adaptive and user-centered robot behavior.
- Design, execution, and validation of Human-Robot Interaction (HRI) experiments with healthy subjects and patients, integrating multimodal perception and adaptive control.

EDUCATION AND TRAINING

- 2019 – 2022 **Doctor of Philosophy in Science and Engineering for Humans and the Environments: Curriculum Bioengineering (ING-IND/34)**
- Thesis title: *"User State Estimation Methods for Tailoring Human-Robot Interaction"*
 - The thesis proposes innovative methods to improve robot-assisted rehabilitation and worker health, including adapted control strategies and the use of wearable sensors. It also explores the positive impact of collaborative robots and wearable passive exoskeletons, providing innovative solutions in contexts such as industry and COVID centers.
 - Supervisor: Prof. Loredana Zollo - CREO Lab, Università Campus Bio-Medico di Roma
Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy
- 2020 **State certification to practice as an engineer**
- Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy
- 2019 – 2020 **Postgraduate Advanced School in Artificial Intelligence**
- The School is dedicated to training specialized professionals and researchers in AI, particularly deep neural networks, and computational modeling of the brain, behavior, and society. While not exhaustive in covering all AI topics, these areas serve as a foundational focus, building on the core competencies of the supporting research and professional network.
 - Main Courses: Machine Learning, Deep Learning, Computer Vision
ISTC-CNR Roma 1 (sede centrale) Via Giandomenico Romagnosi 18a, 00196, Roma (RM), Italy
- 2016 – 2018 **M.Sc. in Biomedical Engineering (LM-21)**
- Thesis Title: *"Trajectory Planning of working tasks and control of an anthropomorphic manipulator with the integrated use of DMP and RGB-D camera"*
 - Mark 110/110 cum laude
 - This thesis proposes a robotic rehabilitation system for patients with musculoskeletal disorders in working environments. Using an RGB-D camera and an anthropomorphic manipulator, modules were developed to estimate the pose of objects, plan trajectories, and create a database of working gestures.
 - Supervisor: Prof. Loredana Zollo - CREO Lab, Università Campus Bio-Medico di Roma
 - Co-Supervisors: Eng. Francesca Cordella and Eng. Clemente Lauretti - CREO Lab, Università Campus Bio-Medico di Roma
Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy
- 2012 – 2016 **B.Sc. in Medical Engineering (L-9)**
- Mark 102/110
University of Tor Vergata, Via Cracovia, 50, 00133, Rome (RM), Italy
- 2006 – 2012 **High school leaving qualification in scientific studies**
- Mark 82/100
Scientific High School "Paolo Ruffini", Piazza Dante Alighieri, 13, 01100, Viterbo (VT), Italy

WORKING EXPERIENCE

May 2024 – current **Fixed Term Researcher**

I am responsible for developing and validating automatic systems for motor and cognitive rehabilitation using social robotic systems. This involves the creation of a cognitive architecture for the planning of customized exercise programmes and the provision of real-time feedback.

Institute of Cognitive Science and Technology – ISTC-CNR, Via Giandomenico Romagnosi 18a, 00196, Roma (RM), Italy

Feb 2023 – May 2024 **Postdoctoral Research Fellow**

I have been awarded a research grant to conduct a research collaboration on the "**SPINE4.0**: Development of an innovative, multi-disciplinary and integrated approach for workers suffering from degenerative pathologies of the lumbar spine based on advanced technologies, capacity building and feasibility analysis for the creation of a reference center for prevention, diagnosis, treatment and reintegration into work" project with the Advanced Robotics and Human-Centred Technologies (CREO Lab)

Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy

2023 Expert Reviewer of research projects for Hungarian National Research

Development and Innovation Office, Hungary

Nov 2019 – Nov 2022 Ph.D. Scholarship

I have been awarded a scholarship to pursue my **Ph.D. program in Science and Engineering for Humans and the Environments**: Curriculum Bioengineering (ING-IND/34), XXXV Cycle - A.Y. 2019-2020, at Università Campus Bio-Medico di Roma.

Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy

Jan 2019 – Feb 2023 Collaboration contracts

Research collaboration with the Advanced Robotics and Human-Centred Technologies (CREO Lab) on the following research projects:

Nov 2022 – Feb 2023 **ODIN**: Leveraging AI based technology to transform the future of health care delivery in Leading Hospitals in Europe

Sep 2019 – Oct 2019 **SENSE-RISC**: Development of instrumented suits for prevention and mitigation of workers' safety risks

Jul 2019 – Aug 2019 **SAFE-MOVER**: User-centred design of a robotic device for improving working conditions and user subjective perspective during patient-handling movements

Feb 2019 – Jun 2019 **RehabRobo@work**: Bio-cooperative robotic system for upper-limb rehabilitation in working environments

Jan 2019 **SIRASI**: Robotic system for upper and lower limb rehabilitation

Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy

FOREIGN EXPERIENCES

Jun 2022 – Nov 2022 Visiting student

- Finalist at the Kuka Innovation Award 2022 with the "SAFER" Team, video at [YouTube](#).
- *Role*: development of user state estimation algorithms to predict the active level of participation of healthy subjects undergoing robot-aided rehabilitation sessions.
- *Publication*: the work carried out during abroad was published in the Q1-ranked journal *Frontiers in Digital Health* [15].
- Supervisor: Prof. Bram Vanderborght.

Vrije Universiteit Brussel, Bd de la Plaine, 2, 1050, Brussel (BRU), Belgium

EDUCATIONAL ACTIVITIES

Lecturer *"Cyber Physical-Robotics"*, 2nd year, MSc in Intelligent Systems Engineering

– A.Y. 23/24 2/9 CFU

Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy

"Python, Basic programming tools (Anaconda, Colab)", Advanced School in Artificial Intelligence, ISTC - CNR since the 7th edition

ISTC Roma 1 (sede centrale) Via Giandomenico Romagnosi 18a, 00196, Roma (RM), Italy

"AI Lab: ML vs deep network for motion perception", Advanced School in Artificial Intelligence, ISTC - CNR since the 2nd edition

ISTC Roma 1 (sede centrale) Via Giandomenico Romagnosi 18a, 00196, Roma (RM), Italy

- Modulo didattico di "Programmazione e Controllo di Robot" nell'ambito del Corso "Tecnico Superiore per l'automazione ed i Sistemi mecatronici Specialista per la transizione Digitale dell'industria Chimico-Farmaceutica" presso la Fondazione ITS Meccatronico del Lazio
- Fondazione ITS Meccatronico del Lazio, Via del Plebiscito, 15, 03100, Frosinone (FR), Italy
- Teaching assistant *"Industrial and Medical Robotics"*, 1st year, MSc in Biomedical Engineering
- A.Y. 25/26, 5/15 CFU
 - A.Y. 24/25, 5/15 CFU
 - A.Y. 23/24, 8/15 CFU
 - A.Y. 22/23, 7.5/15 CFU
 - A.Y. 21/22, 7.5/15 CFU
 - A.Y. 20/21, 6/15 CFU
 - A.Y. 19/20, 3/15 CFU
- "Bioengineering and biomechanics of Human Motions"*, 2nd year, MSc in Biomedical Engineering
- A.Y. 23/24, 4/6 CFU
 - A.Y. 22/23, 5/6 CFU
 - A.Y. 21/22, 5/6 CFU
 - A.Y. 20/21, 3/6 CFU
- Università Campus Bio-Medico di Roma, Via Álvaro del Portillo, 21, 00128, Rome (RM), Italy
- Other
- Co-supervisor of two Ph.D. students (National Ph.D. in Artificial Intelligence, XXXVIII-bis cycle, Health and Life Sciences track, Università Campus Bio-Medico di Roma), contributing to the definition and scientific development of two complementary research lines:
 - Omar Coser - **Lower-limb robot-aided rehabilitation and AI**: methodological work on the integration of machine learning and deep learning models for human motion analysis and user-state estimation. Activities led to:
 - a systematic review [17] regarding the AI methodologies applied for lower-limb rehabilitation robotics, awarded *Best Paper 2024* by *Frontiers in Robotics and AI*;
 - a comparative study of deep learning methodologies for gait-related parameter extraction [6], published in *Frontiers in Computer Science*.
 - Rita Molle - **Upper-limb robot-aided rehabilitation and adaptive personalization**: the work focuses on modelling expert physiotherapists' decision strategies to drive intelligent personalization of upper-limb treatment. Activities led to:
 - a systematic review assessing the AI methodologies applied for upper-limb robot-aided physical rehabilitation [9] published in *ACM Transactions on Human-Robot Interaction*.
 - a study to evaluate therapists' decision-making during robot-mediated rehabilitation sessions [7], published in *Int. Journal of Social Robotics*.
 - a Reinforcement-Learning-based methodology to automatically tune the control parameters of a rehabilitation robot during the therapy [8], published in the prestigious *Engineering Applications of Artificial Intelligence* journal.
 - Co-supervisor of:
 - 15 Bachelor theses in Industrial Engineering, Università Campus Bio-Medico di Roma
 - 13 Master theses in Biomedical Engineering, Università Campus Bio-Medico di Roma
 - 1 Master thesis in Computer Science, Università Federico II di Napoli
 - Personal tutoring activity: supporting first- and second-year Industrial Engineering students in managing their academic experience at Università Campus Bio-Medico di Roma.

SKILLS AND COMPETENCES

Technical Skills

- Extensive experience in Human-Robot Interaction (HRI) experiments, including multimodal data acquisition and analysis (physiological, visual, and kinematic), and in the design of real-time classification and user-state estimation algorithms for adaptive and user-tailored robot behavior.
- Proficient in image processing and computer vision for gesture, facial expression, action, and object recognition using OpenCV, MediaPipe, and deep learning frameworks.
- Advanced programming skills in Python, Java, MATLAB, C#, and HTML, with practical experience in robotic middleware such as ROS and KUKA Sunrise.OS.
- Expertise in robot interfacing, planning, and control for various robotic platforms (Kuka LWR 4+, Kuka LBR Med, TIAGo).
- Experience in data acquisition, sensor integration, and post-processing pipelines for robotic and rehabilitation systems.
- Skilled in the development of Virtual Reality environments for rehabilitation and training applications using Unity (C#).

Soft Skills

- **Teaching:** Experience instructing students through engaging lessons.
- **Keenly Curious:** Exhibits a keen curiosity, consistently seeking to broaden knowledge and explore new concepts.
- **Problem-Solving:** Demonstrated ability to tackle complex challenges through critical analysis and finding innovative solutions.
- **Team Working:** Effective collaboration with colleagues during academic and professional projects.
- **Motivation:** Ability to inspire and motivate others towards goal achievement.
- **Flexibility:** Adaptation to changing needs and priorities in a dynamic environment.
- **Perseverance:** Unwavering commitment and persistence in overcoming obstacles and challenges.
- **Cross-functional communication:** Demonstrated proficiency in effectively communicating with professionals spanning diverse fields, including clinical staff such as surgeons, doctors, and physiotherapists, as well as technical experts and end-users of the technology (i.e., patients).

Languages

Mother tongue Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	B2	B2	B2	B2	B2

JOURNAL AND CONFERENCE SERVICE

Editorial Roles for International Journals	Associate Editor for Discover Robotics, Springer Nature .
	Associate Editor for Discover Artificial Intelligence, Springer Nature .
	Associate Editor for Intelligenza Artificiale, Sage Journals .
	Associate Editor for Frontiers in Digital Health, Connected Health Section .
Editorial Roles for International Conferences	Review Editor for Frontiers in Robotics and AI, Biomedical Robotics Section .
	Associate Editor for the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) since 2026.

Guest Editor of Special Issue	Guest Editor of the Special Issue on <i>Robotics and Emerging Technologies in Rehabilitation: The Robot as an Assistive Tool for Both Assessment and Personalized Intervention</i> Frontiers on Robotics and AI. Topic Coordinator of the Research Topic on <i>Advancements in AI-driven Multimodal Interfaces for Robot-Aided Rehabilitation</i> on Frontiers on Robotics and AI, Frontiers on Big Data, and Frontiers on Artificial Intelligence Journals. Guest Editor of the Special Issue on <i>Artificial Intelligence and Intelligent Robots: Challenges and Opportunities</i> on Applied Sciences MDPI.
Conference Chair Roles	Member of the Organizing Committee of the <i>Robotics and Emerging Technologies in Rehabilitation: The Robot as a Tool for Both Assessment and Personalized Intervention</i> workshop held in conjunction with the Italian Institute of Robotics and Intelligent Machines 3D Conference (I-RIM 2025). Member of the Local Organizing Committee of the <i>14th Forum of Italian Ambient Assisted Living (FortAAL 2025)</i>. Member of the Organizing Committee of the <i>Workshop on Social Robotics for Human-Centered Assistive and Rehabilitation AI</i> held in conjunction with the ICSR 2025 international conference. Program Chair of the <i>Workshop on Advanced AI Methods and Interfaces for Human-Centered Assistive and Rehabilitation Robotics (a Fit4MedRob event) - AlxIA 2024</i> Program Chair of the <i>ALTRUIST 4th Workshop on "sociAL roboTs for peRsonalized, continUous and adaptlve aSsisTance"</i> Program Chair of the <i>AlxIA 2024 (23rd International Conference of the Italian Association for Artificial Intelligence)</i> Travel Grant Chair of the <i>AlxIA 2024 (23rd International Conference of the Italian Association for Artificial Intelligence)</i> Member of the Local Organizing Committee of the <i>Sixth Italian Conference on Robotics and Intelligent Machines</i>
Reviewer for International Journals	IEEE: Transactions on Human-Machine Systems, Sensors Journal, Robotics and Automation Letters, Transactions on Vehicular Technology, Transactions on Medical Robotics and Bionics, Access, Transactions on Haptics, Transactions on Robotics. Wiley: Applied Bionics and Biomechanics Sage Journals: International Journal of Advanced Robotic Systems, Prosthetics and Orthotics International. Taylor and Francis: Computer Methods in Biomechanics and Biomedical Engineering, Disability and Rehabilitation: Assistive Technology. Elsevier: Expert Systems with Applications. Springer Nature: Scientific Reports, SN Applied Sciences, npj Digital Medicine Frontiers: Frontiers in Neurorobotics, Frontiers in Robotics and AI, Frontiers in Computer Science, Frontiers in Bioengineering and Biotechnology, Frontiers in Rehabilitation Sciences. MDPI: Future Internet, Life, Robotics, Electronics, Machines, Applied Sciences, Clocks & Sleep, Sensors, Multimodal Technologies and Interaction, Algorithms, Actuators, Mathematics, Journal of Low Power Electronics, Applications, International Journal of Environmental Research and Public Health, Healthcare, Aerospace
Reviewer for International Conferences	IEEE International Conference on Robotics and Automation, IEEE International Conference on Robot & Human Interactive Communication, IEEE International Symposium on Safety, Security, and Rescue Robotics, Sensors and Application Conference 2020, IEEE International Workshop On Metrology for Industry 4.0 & IoT, International Conference of the IEEE Engineering in Medicine and Biology Society, International Conference of the Italian Association for Artificial Intelligence, International Conference on Social Robotics
Reviewer for National Conferences	Congresso Nazionale di Bioingegneria (GNB), Italian Conference on Robotics and Intelligent Machines (I-RIM).
Member of Scientific Societies	– Member of the IEEE, IEEE Robotics and Automation Society, since 2019 – National Group of Bioengineering (GNB), since 2019 – Italian Institute of Robotics and Intelligent Machines (I-RIM), 2020 – Italian Association for Artificial Intelligence (AlxIA), since 2024

ATTENDED CONFERENCES

- Oral presentation at conference
- Oral presentation at the International Conference on Social Robotics + AI (ICSR+AI 25) of the paper "A Multimodal Emotion Recognition Approach for Socially Assistive Robots", 11/09/2025.
 - Oral presentation at the International Conference on Robot and Human Interactive Communication (RO-MAN 2025) conference of the paper "Enhancing Adaptive Robotic Coaches with Multimodal Workload Estimation", 26/08/2025.
 - Oral presentation at the *Workshop on Advanced AI Methods and Interfaces for Human-Centered Assistive and Rehabilitation Robotics (a Fit4MedRob event)* held in parallel with the International Conference of the Italian Association for Artificial Intelligence (AlxIA 2024) of the paper "Leveraging Multimodal Monitoring in Plan-Based Robot-Aided Rehabilitation", 27/11/2024.
 - Oral presentation at the main track of the International Conference of the Italian Association for Artificial Intelligence (AlxIA 2024) of the paper "REPAIR Platform: Robot-AidEd Person-Allzed Rehabilitation", 27/11/2024.
 - Speaker at the "Rehabilitative and Assistive Robotics Research in Italy: The Fit4Medical Robotics Initiative" Workshop, in the framework of the Sixth Italian Conference on Robotics and Intelligent Machines (I-RIM 2024), "Fostering Behavior Change through Cognitive Social Robotics", 25/10/2024.
 - Oral presentation at the ASIMOV workshop Adaptive Social Interaction and MOVement for assistive and rehabilitation robotics, in the framework of the International Conference on Social Robotics (ICSR 2022) of the paper "Psychophysiological Assessment during Upper-Limb Robot-Aided Rehabilitation in Patients with Musculoskeletal Disorders", 12/11/2022.
 - Oral presentation at the National Congress of the Italian Society of Ergonomics and Human Factors of the paper "Ergonomic evaluation of working gestures through an optoelectronic system", 04/05/2022.
 - Oral presentation at the International Conference on Intelligent Robots and Systems (IROS 2021) of the paper "A robotic health-care assistant for COVID-19 emergency", 28/09/2021.
 - Oral presentation at the Congress of the National Group of Bioengineering (GNB 2020) of the paper "A Dataset of DMPs for robot motion planning", 11/06/2021.
 - Oral presentation at the IEEE MetroInd4.0&IoT conference of the paper "Analysis of Physiological Parameters and Workload during Working Tasks in COVID-19 Pandemic Conditions", 08/06/2021.
 - Oral presentation at the Italian Institute of Robotics and Intelligent Machines 3D Conference (I-RIM 2020) conference of the papers "Combined use of DMP and real objects in robot-aided rehabilitation" and "A Robotic Assistant for Logistics and Disinfection in Health Centers", 11/12/2020.
 - Oral presentation at the International Conference on NeuroRehabilitation (ICNR) conference of the paper "A robot-aided rehabilitation platform for occupational therapy with real objects", 16/10/2020.
 - Oral presentation at the BAILAR Behavior Adaptation, Interaction and Learning for Assistive Robotics workshop, in the framework of the International Conference on Robot and Human Interactive Communication (RO-MAN) of the paper "A robot-aided rehabilitation system based on the combined use of Dynamic Motion Primitives and RGB-D camera", 03/09/2020.
- Chair at conference
- Chair of the "*Medical Robots and Systems*" session at the IROS conference, 09/28/2021.

BIBLIOMETRICS

Web Of Science	– Documents: 61	Citations: 354	h-index: 11
Scopus	– Documents: 58	Citations: 484	h-index: 13
ResearchGate	– Documents: 73	Citations: 568	h-index: 14
Google Scholar	– Documents: 73	Citations: 648	h-index: 15

PROJECTS PARTICIPATION

Fit4MedRob: Fit for medical robotics: 44 mesi per rivoluzionare i modelli assistivi e riabilitativi (Project funded by the Italian Ministry of Research, under the complementary actions to the NRRP, PNC0000007)

Role: design and development of intelligent rehabilitation robots for personalized treatment delivery, with a particular emphasis on reactive adaptation to the patient's state and long-term personalization.

Pubblications: main scientific outputs derived from the project [4, 16, 12, 15, 7, 8, 17, 6, 9]

SPINE4.0: Development of an innovative, multi-disciplinary and integrated approach for workers suffering from degenerative pathologies of the lumbar spine based on advanced technologies, capacity building and feasibility analysis for the creation of a reference center for prevention, diagnosis, treatment and reintegration into work. (Project funded by National Institute for Insurance against Accidents at Work, CUP: C85F21001020001)

Role: Reproduction of working scenarios through a virtual reality system and mechatronic and sensory technologies to generate modular loads to assess patients and workers suffering from low back pain.

ODIN: Leveraging AI based technology to transform the future of health care delivery in Leading Hospitals in Europe (Project Founded by European Union's Horizon 2020 Research and Innovation Programme, CUP: C85F21000670006)

Role: Development and validation of vision systems for patient posture recognition for monitoring meal consumption and joint mobilization activity of upper and lower limbs.

Pubblications: main scientific outputs derived from the project [18]

HeAL9000: Healthcare Agents and Learning robots (Project Founded by Regione Lazio, CUP: B84I20001880002)

Role: Design of interaction modalities for a collaborative robot to provide patients with robot-aided rehabilitation treatment. I implemented a monitoring system to retrieve the complete user state regarding facial expression, psychophysiological state, and kinematics performance.

Pubblications: main scientific outputs derived from the project [28]

EXPERIENCE: Benchmarking Exoskeleton-Assisted Gait Based on Users' Subjective Perspective and Experience (Open Call of the EU-funded project EUROBENCH H2020 ICT-2016-2017-779963)

Role: Design of a methodology to assess psychophysiological indicators of patients during exoskeleton-assisted walking. A Fuzzy Logic approach estimating attention, stress, energy expenditure, and fatigue levels was designed and validated on healthy and pathological participants.

Pubblications: main scientific outputs derived from the project [20]

SENSE-RISC: Development of instrumented suits for prevention and mitigation of workers' safety risks (Project funded by National Institute for Insurance against Accidents at Work, CUP: B56C18004200005)

Role: Design and validation of classification and estimation models assessing the activity level and a multiparametric indicator of workers. Integration of the cloud architecture implemented to collect data from wearable sensors with the user state estimation software.

Pubblications: main scientific outputs derived from the project [27, 13]

TOTEM: TOTAl Ergonomic Measurement Platform (Project Horizon 2020 – PON 2014/2020)

Role: Validation of M-IMU sensors by using an anthropomorphic manipulator. The robotic arm was controlled to generate controlled motions to compute M-IMUs tracking accuracy.

Pubblications: main scientific outputs derived from the project [30]

SAFE-MOVER: User-centred design of a robotic device for improving working conditions and user subjective perspective during patient-handling movements (Bando University Strategic Projects, Topic: Healthcare 4.0)

Role: Biomechanical analysis of healthcare operators during patient-handling movements and patients' psychophysiological assessment.

ARONA: Surgical navigation assisted by advanced robotics (Project MIUR PON Research and Innovation 2014 – 2020, CUP: B56C18004200005)

Role: Validation of a hands-on approach for robot-aided pedicle screw fixation. Biomechanical evaluation of participants performing the robot-aided pedicle screw fixation in different tapping modalities.

Publications: main scientific outputs derived from the project [33]

RehabRobo@work: Bio-cooperative robotic system for upper-limb rehabilitation in working environments (Project funded by National Institute for Insurance against Accidents at Work, CUP: E58C13000990001)

Role: Integration of the bio-cooperative robotic platform. Development and validation of a patient-tailored control strategy for end-effector robots. Clinical validation of the platform on patients affected by musculoskeletal disorders.

Publications: main scientific outputs derived from the project [31, 26, 19]

SIRASI: Robotic system for upper and lower limb rehabilitation (Bando INTESE)

Role: Assembly of the modules making up the SIRASI platform and validation employing empty and load handling tests. The robotic platform was validated for both upper and lower limbs.

AWARDS

- 2026 Finalist (Top 13) at the Most Promising Researcher in Robotics and Artificial Intelligence Award as part of the RomeCup 2026. 
- 2025 Finalist (Top 3) for the Best Paper Award at the I-RIM 3D 2025 conference held in Rome with the paper "Identifying Key Parameters in Physiotherapists' Decision-Making for Robot-Aided Rehabilitation".
- 2025 Best Paper Award at the 14th Forum of Italian Ambient Assisted Living (ForItAAL 2025), Rome, for the paper "*DeepPhysioNet: A Deep Physiological Feature Extraction Method for Affective State Recognition from Wearable Sensing*". 
- 2025 Finalist (Top 11) at the Most Promising Researcher in Robotics and Artificial Intelligence Award as part of the RomeCup 2025.  
- 2024 Best Paper Award at the 23rd International Conference of the Italian Association for Artificial Intelligence (AlxIA 2024) conference held in Bolzano with the paper "REPAIR Platform: Robot-AidEd PersonAllzed Rehabilitation" 
- 2024 Supervisor of the winner of the Master Thesis Award, Istituto di Biorobotica of Scuola Superiore Sant'Anna by Gruppo Nazionale di Bioingegneria (Gianmarco Cirelli) for *his excellent results obtained on the semiautonomous control strategy for a hand-wrist prosthesis based on computer vision*.  [21, 2, 1]
- 2020 First place in the sixth "Premio Qualità" competition organized by the Clinical Directorate of the University Hospital Università Campus Bio-Medico di Roma. The Award aimed at presenting and rewarding experiences that, by highlighting virtuous behavior, documented an improvement in quality through an organization of work that put the patient and the operator at the center of its work. 

JOURNAL PAPERS

- [J1] G. Cirelli, E. Stefanelli, R. Billardello, **C. Tamantini**, S. Leonelli, L. Zollo, L. P. Cordella, and F. Cordella. "Context-Aware Semiautonomous Control for Upper-Limb Prostheses". In: *Advanced Intelligent Systems* (2026), e202501043. DOI: [10.1002/aisy.202501043](https://doi.org/10.1002/aisy.202501043).
- [J2] G. Cirelli, **C. Tamantini**, L. Zollo, and F. Cordella. "Computer Vision for High-Level Control of Prosthetic Limbs: A Literature Review". In: *Computer Vision and Image Understanding* (2026). DOI: [10.1016/j.cviu.2026.104669](https://doi.org/10.1016/j.cviu.2026.104669).
- [J3] R. Molle, **C. Tamantini**, C. Lauretti, D. Sebastiani, F. Santacaterina, M. Bravi, F. Bressi, S. Miccinilli, and L. Zollo. "Behavioral Cloning of Physiotherapists in Adapting Robot Control Parameter". In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2026). DOI: [10.1109/TNSRE.2026.3659215](https://doi.org/10.1109/TNSRE.2026.3659215).
- [J4] **C. Tamantini**, A. Umbrico, F. Fracasso, G. Beraldo, G. Cortellessa, and A. Orlandini. "Adapting Virtual Agent Interaction Style with Reinforcement Learning to Enhance Affective Engagement". In: *Frontiers in Digital Health* 7 (2026), p. 1680605. DOI: [10.3389/fdgth.2025.1680605](https://doi.org/10.3389/fdgth.2025.1680605).
- [J5] R. Billardello, **C. Tamantini**, F. Cordella, F. Scotto di Luzio, T. Varrecchia, G. Chini, F. Draicchio, A. Ranavolo, and L. Zollo. "Estimating workers' physical effort during isometric contractions through sEMG: the role of feature selection". In: *ACM Transactions on Human-Robot Interaction* (2025). DOI: [10.1145/3759159](https://doi.org/10.1145/3759159).
- [J6] O. Coser, **C. Tamantini**, M. Tortora, L. Furia, R. Sicilia, L. Zollo, and P. Soda. "Deep learning for human locomotion analysis in lower-limb exoskeletons: a comparative study". In: *Frontiers in Computer Science* 7 (2025). DOI: [10.3389/fcomp.2025.1597143](https://doi.org/10.3389/fcomp.2025.1597143).
- [J7] R. Molle, **C. Tamantini**, C. Lauretti, F. Cordella, F. Scotto di Luzio, D. Sebastiani, F. Santacaterina, M. Bravi, F. Bressi, S. Miccinilli, and L. Zollo. "Exploring Priority Parameters in Physiotherapist Decision Models for Tailoring Robot-Aided Rehabilitation". In: *International Journal of Social Robotics* (2025). DOI: [10.1007/s12369-025-01309-3](https://doi.org/10.1007/s12369-025-01309-3).
- [J8] R. Molle, **C. Tamantini**, C. Lauretti, E. M. Romano, and L. Zollo. "An online reinforcement learning method to improve control adaptability in robot-aided rehabilitation". In: *Engineering Applications of Artificial Intelligence* 161 (2025). DOI: [10.1016/j.engappai.2025.112248](https://doi.org/10.1016/j.engappai.2025.112248).
- [J9] R. Molle, **C. Tamantini**, and L. Zollo. "Artificial Intelligence in Upper Limb Robot-Aided Physical Rehabilitation: a Systematic Review". In: *ACM Transactions on Human-Robot Interaction* (2025). DOI: [10.1145/3779302](https://doi.org/10.1145/3779302).
- [J10] F. Santacaterina, **C. Tamantini**, G. Camarro, S. Miccinilli, F. Bressi, L. Zollo, S. Sterzi, and M. Bravi. "Is Brazilian Jiu-Jitsu a Traumatic Sport? Survey on Italian Athletes' Rehabilitation and Return to Sport". In: *Journal of Functional Morphology and Kinesiology* 10.3 (2025). DOI: [10.3390/jfmk10030286](https://doi.org/10.3390/jfmk10030286).
- [J11] F. Scotto di Luzio, **C. Tamantini**, R. Di Maro, C. Carnazzo, S. Spada, F. Draicchio, and L. Zollo. "Biomechanical and physiological effects of passive upper limb exoskeletons in simulated manufacturing tasks". In: *Wearable Technologies* 6 (2025). DOI: [10.1017/wtc.2025.10021](https://doi.org/10.1017/wtc.2025.10021).
- [J12] **C. Tamantini**, M. L. Cristofanelli, F. Fracasso, A. Umbrico, G. Cortellessa, A. Orlandini, and F. Cordella. "Physiological Sensor Technologies in Workload Estimation: A Review". In: *IEEE Sensors Journal* 25.18 (2025), pp. 34298–34310. DOI: [10.1109/JSEN.2025.3597329](https://doi.org/10.1109/JSEN.2025.3597329).
- [J13] **C. Tamantini**, F. Marra, J. Di Tocco, S. Di Modica, A. Lanatà, F. Cordella, M. Ferrarin, F. Rizzo, M. Stefanelli, M. Papacchini, C. Delle Site, A. Tamburrano, C. Massaroni, E. Schena, L. Zollo, and M. S. Sarto. "SenseRisc: An instrumented smart shirt for risk prevention in the workplace". In: *Wearable Technologies* 6 (2025). DOI: [10.1017/wtc.2025.10](https://doi.org/10.1017/wtc.2025.10).
- [J14] **C. Tamantini**, K. Patrice Langlois, D. R. Cianca, and L. Zollo. "Editorial: Advancements in AI-driven multimodal interfaces for robot-aided rehabilitation". In: *Frontiers in Robotics and AI* 12 (2025). DOI: [10.3389/frobt.2025.1605418](https://doi.org/10.3389/frobt.2025.1605418).
- [J15] **C. Tamantini**, K. Patrice Langlois, J. de Winter, P. H. Ali Mohamadi, D. Beckwée, E. Swinnen, T. Verstraten, B. Vanderborght, and L. Zollo. "Promoting active participation in robot-aided rehabilitation via machine learning and impedance control". In: *Frontiers in Digital Health* 7 (2025). DOI: [10.3389/fdgth.2025.1559796](https://doi.org/10.3389/fdgth.2025.1559796).
- [J16] **C. Tamantini**, A. Umbrico, and A. Orlandini. "Automated planning and scheduling in robot-aided rehabilitation: a review". In: *Journal of NeuroEngineering and Rehabilitation* 22.1 (2025). DOI: [10.1186/s12984-025-01710-z](https://doi.org/10.1186/s12984-025-01710-z).
- [J17] O. Coser, **C. Tamantini**, P. Soda, and L. Zollo. "AI-based methodologies for exoskeleton-assisted rehabilitation of the lower limb: a review". In: *Frontiers in Robotics and AI* 11 (2024). DOI: [10.3389/frobt.2024.1341580](https://doi.org/10.3389/frobt.2024.1341580).

- [J18] C. Rondoni, F. Scotto di Luzio, **C. Tamantini**, N. L. Tagliamonte, M. Chiurazzi, G. Ciuti, and L. Zollo. "Navigation benchmarking for autonomous mobile robots in hospital environment". In: *Scientific Reports* 14.1 (2024). DOI: [10.1038/s41598-024-69040-z](https://doi.org/10.1038/s41598-024-69040-z).
- [J19] **C. Tamantini**, F. Cordella, F. Scotto di Luzio, C. Lauretti, B. Campagnola, F. Santacaterina, M. Bravi, F. Bressi, F. Draicchio, S. Miccinilli, and L. Zollo. "A fuzzy-logic approach for longitudinal assessment of patients' psychophysiological state: an application to upper-limb orthopedic robot-aided rehabilitation". In: *Journal of NeuroEngineering and Rehabilitation* 21.1 (2024). DOI: [10.1186/s12984-024-01501-y](https://doi.org/10.1186/s12984-024-01501-y).
- [J20] **C. Tamantini**, F. Cordella, N. L. Tagliamonte, I. Pecoraro, I. Pisotta, A. Bigioni, F. Tamburella, M. Lorusso, M. Molinari, and L. Zollo. "A Data-Driven Fuzzy Logic Method for Psychophysiological Assessment: An Application to Exoskeleton-Assisted Walking". In: *IEEE Transactions on Medical Robotics and Bionics* (2024), pp. 1–1. DOI: [10.1109/TMRB.2024.3377453](https://doi.org/10.1109/TMRB.2024.3377453).
- [J21] G. Cirelli, **C. Tamantini**, L. P. Cordella, and F. Cordella. "A Semiautonomous Control Strategy Based on Computer Vision for a Hand–Wrist Prosthesis". In: *Robotics* 12.6 (2023). DOI: [10.3390/robotics12060152](https://doi.org/10.3390/robotics12060152).
- [J22] R. Cittadini, **C. Tamantini**, F. Scotto di Luzio, C. Lauretti, L. Zollo, and F. Cordella. "Affective state estimation based on Russell's model and physiological measurements". In: *Scientific Reports* 13.1 (2023). DOI: [10.1038/s41598-023-36915-6](https://doi.org/10.1038/s41598-023-36915-6).
- [J23] C. Lauretti, **C. Tamantini**, H. Tomè, and L. Zollo. "Robot Learning by Demonstration with Dynamic Parameterization of the Orientation: An Application to Agricultural Activities". In: *Robotics* 12.6 (2023). DOI: [10.3390/robotics12060166](https://doi.org/10.3390/robotics12060166).
- [J24] C. Lauretti, **C. Tamantini**, and L. Zollo. "A new DMP scaling method for robot learning by demonstration and application to the agricultural domain". In: *IEEE Access* (2023), pp. 1–1. DOI: [10.1109/ACCESS.2023.3349093](https://doi.org/10.1109/ACCESS.2023.3349093).
- [J25] C. Lauretti, **C. Tamantini**, A. Zompanti, S. Cimini, L. De Gara, M. Santonico, and L. Zollo. "A low-cost multispectral device for in-field fruit ripening assessment". In: *IEEE Sensors Journal* (2023), pp. 1–1. DOI: [10.1109/JSEN.2023.3335259](https://doi.org/10.1109/JSEN.2023.3335259).
- [J26] **C. Tamantini**, F. Cordella, C. Lauretti, F. Scotto di Luzio, B. Campagnola, L. Cricenti, M. Bravi, F. Bressi, F. Draicchio, S. Sterzi, and L. Zollo. "Tailoring Upper-Limb Robot-Aided Orthopedic Rehabilitation on Patients' Psychophysiological State". In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2023), pp. 3297–3306. DOI: [10.1109/TNSRE.2023.3298381](https://doi.org/10.1109/TNSRE.2023.3298381).
- [J27] **C. Tamantini**, C. Rondoni, F. Cordella, E. Guglielmelli, and L. Zollo. "A Classification Method for Workers' Physical Risk". In: *Sensors* 23.3 (2023). DOI: [10.3390/s23031575](https://doi.org/10.3390/s23031575).
- [J28] **C. Tamantini**, F. Scotto di Luzio, C. Daniel Hromei, L. Cristofori, D. Croce, M. Cammisa, A. Cristofaro, M. V. Marabello, R. Basili, and L. Zollo. "Integrating Physical and Cognitive Interaction Capabilities in a Robot-Aided Rehabilitation Platform". In: *IEEE Systems Journal* (2023), pp. 1–12. DOI: [10.1109/JSYST.2023.3317504](https://doi.org/10.1109/JSYST.2023.3317504).
- [J29] J. M. Vicente-Samper, **C. Tamantini**, E. Ávila-Navarro, M. A. De La Casa-Lillo, L. Zollo, J. M. Sabater-Navarro, and F. Cordella. "An ML-Based Approach to Reconstruct Heart Rate from PPG in Presence of Motion Artifacts". In: *Biosensors* 13.7 (2023). DOI: [10.3390/bios13070718](https://doi.org/10.3390/bios13070718).
- [J30] M. Lapresa, **C. Tamantini**, F. Scotto Di Luzio, M. Ferlazzo, G. Sorrenti, F. Corpina, and L. Zollo. "Validation of Magneto-Inertial Measurement Units for Upper-Limb Motion Analysis Through an Anthropomorphic Robot". In: *IEEE Sensors Journal* 22.17 (2022), pp. 16920–16928. DOI: [10.1109/JSEN.2022.3193313](https://doi.org/10.1109/JSEN.2022.3193313).
- [J31] **C. Tamantini**, F. Cordella, C. Lauretti, and L. Zollo. "The WGD—A dataset of assembly line working gestures for ergonomic analysis and work-related injuries prevention". In: *Sensors* 21.22 (2021). DOI: [10.3390/s21227600](https://doi.org/10.3390/s21227600).
- [J32] **C. Tamantini**, F. Scotto Di Luzio, F. Cordella, G. Pascarella, F. E. Agrò, and L. Zollo. "A Robotic Health-Care Assistant for COVID-19 Emergency: A Proposed Solution for Logistics and Disinfection in a Hospital Environment". In: *IEEE Robotics and Automation Magazine* 28.1 (2021), pp. 71–81. DOI: [10.1109/MRA.2020.3044953](https://doi.org/10.1109/MRA.2020.3044953).
- [J33] C. Lauretti, F. Cordella, **C. Tamantini**, C. Gentile, F. Scotto Di Luzio, and L. Zollo. "A Surgeon-Robot Shared Control for Ergonomic Pedicle Screw Fixation". In: *IEEE Robotics and Automation Letters* 5.2 (2020), pp. 2554–2561. DOI: [10.1109/LRA.2020.2972892](https://doi.org/10.1109/LRA.2020.2972892).

CONFERENCE PAPERS

- [C1] G. Beraldo, **C. Tamantini**, A. Umbrico, and A. Orlandini. "Evaluating Users' Preferences Regarding Technologies for Promoting Behavioral Change". In: *14th Italian Forum of Ambient Assisted Living (ForItAAL 2025)*. Accepted for presentation. 2025.
- [C2] G. Beraldo, **C. Tamantini**, A. Umbrico, and A. Orlandini. "Personalizing Behavior Change Intervention with Cognitive Robotics: Stage Detection & Goal Reasoning". In: *Workshop on Advanced AI Methods and Interfaces for Human-Centered Assistive and Rehabilitation Robotics 2024 (AlxIA 2024 co-located)*. 2025.
- [C3] G. Cirelli, **C. Tamantini**, L. P. Cordella, and F. Cordella. "AI-based Semiautonomous Control Strategy for upper-limb prostheses". In: *11th Italian Workshop on Artificial Intelligence and Robotics (AlxIA 2024 co-located)*. 2025.
- [C4] R. Molle, **C. Tamantini**, C. Lauretti, F. Cordella, F. Scotto di Luzio, D. Sebastiani, F. Santacaterina, M. Bravi, F. Bressi, S. Miccinilli, and L. Zollo. "Assessing the "Rule of 10" in Orthopedic Robot-Aided Rehabilitation for Tailoring Exercise Dose". In: *Workshop on Advanced AI Methods and Interfaces for Human-Centered Assistive and Rehabilitation Robotics 2024 (AlxIA 2024 co-located)*. 2025.
- [C5] R. Molle, **C. Tamantini**, F. Taffoni, A. Caroppo, A. Manni, P. A. Siciliano, L. Zollo, and A. Leone. "Validating ViSCOPE: Vital Signs Contactless Estimation Pipeline for Robot-Aided Rehabilitation". In: *Workshop on Advanced AI Methods and Interfaces for Human-Centered Assistive and Rehabilitation Robotics 2024 (AlxIA 2024 co-located)*. 2025.
- [C6] F. Rahimi, **C. Tamantini**, A. Orlandini, F. Fracasso, and R. Siciliano. "Comparing Fusion Strategies for Multimodal Emotion Prediction using Deep Physiological Features". In: *Workshop on Social Robotics for Human-Centered Assistive and Rehabilitation AI (Fit4MedRob) – ICSR 2025*. 2025.
- [C7] F. Rahimi, **C. Tamantini**, A. Orlandini, F. Fracasso, and R. Siciliano. "DeepPhysioNet: A Deep Physiological Feature Extraction Method for Affective State Recognition from Wearable Sensing". In: *14th Italian Forum of Ambient Assisted Living (ForItAAL 2025)*. Accepted for presentation. 2025.
- [C8] F. Scotto di Luzio, **C. Tamantini**, C. Lauretti, F. Candeloro, and L. Zollo. "Affective Evaluation of Rehabilitation Tasks Demonstrated by a Service Robot in Joint and Operational Spaces". In: *International Conference on Social Robotics*. Springer. 2025, pp. 607–612. DOI: [10.1007/978-981-95-2382-5_49](https://doi.org/10.1007/978-981-95-2382-5_49).
- [C9] **C. Tamantini**, G. Beraldo, A. Umbrico, and A. Orlandini. "Adaptive Interaction Style Modulation via Reinforcement Learning and Prompted Language Generation". In: *Workshop on Social Robotics for Human-Centered Assistive and Rehabilitation AI (Fit4MedRob) – ICSR 2025*. 2025.
- [C10] **C. Tamantini**, M. L. Cristofanelli, A. Umbrico, F. Fracasso, G. Cortellessa, F. Cordella, and A. Orlandini. "Enhancing Adaptive Robotic Coaches with Multimodal Workload Estimation". In: *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. 2025, pp. 455–461. DOI: [10.1109/RO-MAN63969.2025.11217530](https://doi.org/10.1109/RO-MAN63969.2025.11217530).
- [C11] **C. Tamantini**, M. Fabrizi, L. Zollo, and F. Cordella. "A Multimodal Emotion Recognition Approach for Socially Assistive Robots". In: *International Conference on Social Robotics*. Springer. 2025, pp. 30–41. DOI: [10.1007/978-981-95-2382-5_3](https://doi.org/10.1007/978-981-95-2382-5_3).
- [C12] **C. Tamantini**, A. Umbrico, A. Fabrizio, A. Carnevale, E. Schena, UG. Longo, and A. Orlandini. "Conceptual Framework for Autonomous Coaching in Orthopaedic Rehabilitation with Socially Assistive Robots". In: *International Conference on Social Robotics*. Springer. 2025, pp. 578–583. DOI: [10.1007/978-981-95-2379-5_49](https://doi.org/10.1007/978-981-95-2379-5_49).
- [C13] **C. Tamantini**, A. Umbrico, and A. Orlandini. "Leveraging Multimodal Monitoring in Plan-Based Robot-Aided Rehabilitation". In: *Workshop on Advanced AI Methods and Interfaces for Human-Centered Assistive and Rehabilitation Robotics 2024 (AlxIA 2024 co-located)*. 2025.
- [C14] **C. Tamantini**, A. Umbrico, and A. Orlandini. "REPAIR Platform: Robot-AidEd PersonAllzed Rehabilitation". In: *Lecture Notes in Computer Science*. Vol. 15450. 2025, pp. 301–314. DOI: [10.1007/978-3-031-80607-0_23](https://doi.org/10.1007/978-3-031-80607-0_23).
- [C15] F. Aksu, A. Bria, A. N. Caragliano, C. M. Caruso, W. Chen, E. Cordelli, O. Coser, A. Francesconi, L. Furia, V. Guarrasi, G. Iannello, C. Lauretti, G. Manni, G. Marino, D. Paolo, P. Ruffini, L. Shen, R. Sicilia, P. Soda, **C. Tamantini**, M. Tortora, Z. Wu, and L. Zollo. "Towards AI-driven Next Generation Personalized Healthcare and Well-being". In: *Ital-IA*. 2024.
- [C16] G. Beraldo, **C. Tamantini**, A. Umbrico, and A. Orlandini. "Fostering Behavior Change through Cognitive Social Robotics". In: *ICSR + AI 2024, Odense, Denmark, Oct 23–26*. 2024. DOI: [10.1007/978-981-96-3525-2_24](https://doi.org/10.1007/978-981-96-3525-2_24).

- [C17] C. Lauretti, **C. Tamantini**, A. Zompanti, S. Cimini, L. De Gara, M. Santonico, and L. Zollo. "An Intelligent Robotic Platform for Fruit Selective Harvesting". In: *European Robotics Forum 2024*. Cham: Springer Nature Switzerland, 2024, pp. 252–257. ISBN: 978-3-031-76424-0. DOI: [10.1007/978-3-031-76424-0_45](https://doi.org/10.1007/978-3-031-76424-0_45).
- [C18] R. Molle, **C. Tamantini**, A. Paglialonga, D. Simeone, L. Furia, M. Tortora, C. Lauretti, G. Iannello, P. Soda, R. Sicilia, et al. "AI-CARE: Artificial Intelligence for Customized Adaptive Robot-Aided Rehabilitation". In: *International Workshop on Human-Friendly Robotics*. Springer. 2024, pp. 168–181. DOI: [10.1007/978-3-031-81688-8_13](https://doi.org/10.1007/978-3-031-81688-8_13).
- [C19] F. Scotto di Luzio, **C. Tamantini**, G. Boutaib, C. Carnazzo, S. Spada, and L. Zollo. "Kinematic and Muscular Assessment of an Active Hand Exoskeleton for Industrial Applications". In: *European Robotics Forum 2024*. Cham: Springer Nature Switzerland, 2024, pp. 3–7. ISBN: 978-3-031-76428-8. DOI: [10.1007/978-3-031-76428-8_1](https://doi.org/10.1007/978-3-031-76428-8_1).
- [C20] I. Pecoraro, N. L. Tagliamonte, **C. Tamantini**, F. Cordella, F. Bentivoglio, I. Pisotta, A. Bigioni, F. Tamburella, M. Lorusso, P. Argentieri, M. Molinari, and L. Zollo. "Psychophysiological Assessment of Exoskeleton-Assisted Treadmill Walking". In: *Biosystems and Biorobotics*. Vol. 28. 2022, pp. 201–205. DOI: [10.1007/978-3-030-70316-5_33](https://doi.org/10.1007/978-3-030-70316-5_33).
- [C21] **C. Tamantini**, F. Cordella, C. Lauretti, F. Scotto Di Luzio, M. Bravi, F. Bressi, F. Draicchio, S. Sterzi, and L. Zollo. "Patient-tailored Adaptive Control for Robot-aided Orthopaedic Rehabilitation". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2022, pp. 5434–5440. DOI: [10.1109/ICRA46639.2022.9811791](https://doi.org/10.1109/ICRA46639.2022.9811791).
- [C22] **C. Tamantini**, M. Lapresa, F. Cordella, F. Scotto di Luzio, C. Lauretti, and L. Zollo. "A Robot-Aided Rehabilitation Platform for Occupational Therapy with Real Objects". In: *Biosystems and Biorobotics*. Vol. 28. 2022, pp. 851–855. DOI: [10.1007/978-3-030-70316-5_136](https://doi.org/10.1007/978-3-030-70316-5_136).
- [C23] L. Cristofori, C. D. Hromei, F. Scotto di Luzio, **C. Tamantini**, F. Cordella, D. Croce, L. Zollo, and R. Basili. "HeAL9000: An Intelligent Rehabilitation Robot". In: *CEUR Workshop Proceedings*. Vol. 3060. 2021, pp. 29–41.
- [C24] **C. Tamantini**, M. Lapresa, F. Scotto Di Luzio, F. Cordella, and L. Zollo. "Analysis of physiological parameters and workload during working tasks in COVID-19 pandemic conditions". In: *IEEE International Workshop on Metrology for Industry 4.0 and IoT (MetroInd 4.0 & IoT)*. 2021, pp. 423–428. DOI: [10.1109/MetroInd4.0IoT51437.2021.9488444](https://doi.org/10.1109/MetroInd4.0IoT51437.2021.9488444).
- [C25] A. Lanata, A. Greco, S. Di Modica, F. Niccolini, F. Vivaldi, F. Di Francesco, **C. Tamantini**, F. Cordella, L. Zollo, M. Di Rienzo, C. Massaroni, E. Schena, M. Di Sarto, and E. P. Scilingo. "A New Smart-Fabric based Body Area Sensor Network for Work Risk Assessment". In: *IEEE International Workshop on Metrology for Industry 4.0 and IoT (MetroInd 4.0 & IoT)*. 2020, pp. 187–190. DOI: [10.1109/MetroInd4.0IoT48571.2020.9138273](https://doi.org/10.1109/MetroInd4.0IoT48571.2020.9138273).
- [C26] M. Lapresa, **C. Tamantini**, F. Scotto Di Luzio, F. Cordella, M. Bravi, S. Miccinilli, and L. Zollo. "A Smart Solution for Proprioceptive Rehabilitation through M-IMU Sensors". In: *IEEE International Workshop on Metrology for Industry 4.0 and IoT (MetroInd 4.0 & IoT)*. 2020, pp. 591–595. DOI: [10.1109/MetroInd4.0IoT48571.2020.9138193](https://doi.org/10.1109/MetroInd4.0IoT48571.2020.9138193).
- [C27] **C. Tamantini**, C. Lauretti, F. Cordella, and L. Zollo. "A Dataset of DMPs for robot motion planning". In: *Convegno Nazionale di Bioingegneria*. 2020, pp. 202–205.

WORKSHOP PAPERS

- [W1] R. Molle, **C. Tamantini**, C. Lauretti, D. Sebastiani, and L. Zollo. *Identifying Key Parameters in Physiotherapists' Decision-Making for Robot-Aided Rehabilitation*. VII Robotics and Intelligent Machines Conference (I-RIM 3D 2025), 2025.
- [W2] **C. Tamantini**, G. Beraldo, A. Umbrico, and A. Orlandini. *Empathic Response Generation via LLMs: Measuring Style Differences with Sentiment Metrics*. VII Robotics and Intelligent Machines Conference (I-RIM 3D 2025), 2025.
- [W3] G. Beraldo, **C. Tamantini**, A. Umbrico, and A. Orlandini. *A Dual-Process Cognitive Architecture for Tailored Behavior Change Interventions*. VI Robotics and Intelligent Machines Conference (I-RIM 2024), 2024.
- [W4] G. Cirelli, **C. Tamantini**, E. Stefanelli, R. Billardello, L. Zollo, L. P. Cordella, and F. Cordella. *Usability and Performance Assessment of Semiautonomous Hand-Wrist Prosthetic Control*. VI Robotics and Intelligent Machines Conference (I-RIM 2024), 2024.

- [W5] R. Molle, **C. Tamantini**, C. Lauretti, E. M. Romano, and L. Zollo. *Adapting Interaction Control in Robot-Aided Rehabilitation Using Reinforcement Learning*. VI Robotics and Intelligent Machines Conference (I-RIM 2024), 2024.
- [W6] R. Molle, L. Zollo, and **C. Tamantini**. *Empowering Upper-Limb Robot-Aided Rehabilitation with Artificial Intelligence*. Vol. 18. 1. Proceedings of the MEi: CogSci Conference, 2024.
- [W7] C. Rondoni, F. Scotto di Luzio, **C. Tamantini**, N. L. Tagliamonte, M. Chiurazzi, G. Ciuti, and L. Zollo. *Assessing Robot Navigation in Healthcare: A New Benchmarking Approach for Dynamic Environments*. VI Robotics and Intelligent Machines Conference (I-RIM 2024), 2024.
- [W8] **C. Tamantini**, A. Umbrico, and A. Orlandini. *An Automated Planning Approach for Personalized Robot-Aided Rehabilitation Sessions*. VI Robotics and Intelligent Machines Conference (I-RIM 2024), 2024.
- [W9] R. Billardello, **C. Tamantini**, F. Scotto di Luzio, F. Cordella, F. Russo, G. Papalia, G. Vadalà, and L. Zollo. *Quantifying muscular activity during Back Performance Scale tasks*. XXI Congresso della Società Italiana di Analisi del Movimento in Clinica, 2023.
- [W10] C. Lauretti, **C. Tamantini**, A. Zompanti, S. Cimini, L. De Gara, M. Santonico, and L. Zollo. *A bio-inspired robotic platform for selective harvesting*. Living Machines, 2023.
- [W11] C. Lauretti, **C. Tamantini**, A. Zompanti, S. Cimini, L. De Gara, M. Santonico, and L. Zollo. *A learning-by-demonstration approach for fruit robot-based selective harvesting*. V Robotics and Intelligent Machines Conference (I-RIM 2023), 2023.
- [W12] T. Varrecchia, G. Chini, S. Di Fonzo, R. Billardello, **C. Tamantini**, F. Cordella, F. Scotto di Luzio, L. Zollo, L. Fiori, A. Tatarelli, A. Silvetti, M. Serrao, F. Draicchio, and A. Ranavolo. *Evaluation of a passive upper limb exoskeleton using high-density surface electromyography*. XXI Congresso della Società Italiana di Analisi del Movimento in Clinica, 2023.
- [W13] F. Bressi, Dorotea Manco, S. Miccinilli, M. Bravi, L. Cricenti, B. Campagnola, F. Santacaterina, **C. Tamantini**, F. Scotto di Luzio, F. Cordella, et al. *Sviluppo e validazione di un sistema robotico bio-cooperativo per la riabilitazione ed il reinserimento in contesti lavorativi di soggetti affetti da patologia muscolo-scheletrica acuta del distretto di spalla*. Giornale italiano di Medicina Riabilitativa, 2021.
- [W14] F. Cordella, **C. Tamantini**, F. Scotto di Luzio, B. Albanesi, B. Campagnola, M. Bravi, M. Piredda, F. Bressi, M. G. De Marinis, and L. Zollo. *Kinematic analysis of healthcare workers during patient-handling*. XXI Congresso della Società Italiana di Analisi del Movimento in Clinica, 2021.
- [W15] C. Lauretti, F. Cordella, **C. Tamantini**, C. Gentile, F. Scotto di Luzio, and L. Zollo. *Robot-assisted Procedure for Pedicle Screw Fixation*. III Robotics and Intelligent Machines Conference (I-RIM 2021), 2021.
- [W16] F. Scotto di Luzio, **C. Tamantini**, F. Cordella, C. D. Hromei, L. Cristofori, D. Croce, R. Basili, and L. Zollo. *HeAL9000: an Intelligent Rehabilitation Robot for Physical and Cognitive Interaction*. III Robotics and Intelligent Machines Conference (I-RIM 2021), 2021.
- [W17] **C. Tamantini**, F. Cordella, F. Scotto di Luzio, and L. Zollo. *Psychophysiological Assessment during Upper-Limb Robot-Aided Rehabilitation in Patients with Musculoskeletal Disorders*. ASI-MOV2021: Workshop on Adaptive Social Interaction, MOVement for assistive, and rehabilitation robotics, 2021.
- [W18] I. Pecoraro, N. L. Tagliamonte, **C. Tamantini**, F. Cordella, F. Bentivoglio, I. Pisotta, A. Bigioni, F. Tamburella, M. Lorusso, P. Argentieri, M. Molinari, and L. Zollo. *Assessment of the psychophysiological state during walking with a treadmill-based exoskeleton*. II Robotics and Intelligent Machines Conference (I-RIM 2020), 2020.
- [W19] F. Scotto di Luzio, **C. Tamantini**, G. Boutaib, C. Carnazzo, S. Spada, and L. Zollo. *A Preliminary Ergonomic Evaluation of an Active Hand Exoskeleton for Manufacturing Applications*. XII Congresso nazionale società italiana di ergonomia e fattori umani – SIE2022, 2020.
- [W20] **C. Tamantini**, F. Cordella, F. Scotto di Luzio, C. Lauretti, F. Draicchio, and L. Zollo. *An Ergonomic Evaluation of Working Gestures by means of Optoelectronic Systems*. XII Congresso nazionale società italiana di ergonomia e fattori umani – SIE2022, 2020.
- [W21] **C. Tamantini**, M. Lapresa, F. Cordella, F. Scotto di Luzio, C. Lauretti, and L. Zollo. *A robot-aided rehabilitation system based on the combined use of Dynamic Motion Primitives and RGB-D camera*. Workshop on Behaviors Adaptation, Interaction and Learning for Assistive Robotics (IEEE RO-MAN 2020), 2020.
- [W22] **C. Tamantini**, M. Lapresa, F. Cordella, F. Scotto di Luzio, C. Lauretti, and L. Zollo. *Combined use of DMP and real objects in robot-aided rehabilitation*. II Robotics and Intelligent Machines Conference (I-RIM 2020), 2020.

- [W23] **C. Tamantini**, F. Scotto di Luzio, F. Cordella, M. Lapresa, G. Pascarella, F. E. Agrò, and L. Zollo. *A Robotic Assistant for Logistics and Disinfection in Health Centers*. II Robotics and Intelligent Machines Conference (I-RIM 2020), 2020.

BOOK CHAPTERS

- [B1] C. Lauretti, F. Scotto Di Luzio, A. Demofonti, **C. Tamantini**, F. Cordella, N. L. Tagliamonte, and L. Zollo. "Towards Human-Centric, Sustainable, and Resilient Robot Technologies". In: *Health-care in the Digital Age: Perspectives for Sustainable Innovation and Assessment*. Springer, 2025, pp. 57–87.

In compliance with the GDPR and the Italian Legislative Decree no. 196 dated 30/06/2003, I hereby authorize you to use and process my personal details contained in this document